Standard Operating Procedure for Troubleshooting BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding on Cisco PE-04 at MG Road Core

Node: MG-PE-04 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_5564

Description

BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on Cisco PE-04 at MG Road Core can occur due to a multitude of reasons ranging from network congestion, hardware issues, or misconfigured routing protocols. The PE-04 is equipped with Cisco's IOS-XE software, which utilizes the BGP, OSPF, and ISIS routing protocols for internetworking. BGP route churn can manifest as an excessive number of route advertisements,Withdrawals, or updates, which may lead to routing instability. OSPF adjacency loss may result from network partitions, interface issues, or misconfigured OSPF parameters. Meanwhile, ISIS LSP flooding can occur when the ISIS protocol detects a topology change, leading to a burst of LSP (Link-State Protocol) advertisements. To troubleshoot this failure domain, it is essential to understand the underlying protocol mechanics and utilize the appropriate diagnostic commands to isolate the root cause.

Pre-requisite Checks

- Pre-requisite: Verify that remote hands have been dispatched to MG Road Core for on-site assistance
- Pre-requisite: Check the status of redundant links to ensure redundancy is operational
- Pre-requisite: Review the event history and fault logs to determine the timestamp of the failure

Potential Root Causes

Cause	Probability
BGP route churn due to network congestion	High
OSPF adjacency loss due to network partition	Medium
ISIS LSP flooding due to misconfigured ISIS parameters	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp, Expected Output: BGP table version is 12, main routing table version is 12, Description: This command displays the BGP table version, main routing table version, and other relevant parameters to identify potential issues with BGP route churn.

Command: show ip ospf interface, Expected Output: OSPF interface is in state DOWN, Description: This command displays the OSPF interface state, which can help identify potential issues with OSPF adjacency loss.
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```
Command: show isis database, Expected Output: ISIS LSP flooding detected, Description: This command displays the ISIS LSP database, which can help identify potential issues with ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Clear BGP route churn, Command: clear ip bgp *, Description: This command clears the BGP table and resets the BGP process, which can help resolve BGP route churn issues.

Step 2: Step: Re-establish OSPF adjacency, Command: clear ip ospf process, Description: This command clears the OSPF process and re-establishes OSPF adjacency, which can help resolve OSPF adjacency loss issues.

Step 3: Step: Configure ISIS parameters, Command: router isis, Description: This command configures the ISIS protocol and sets the necessary parameters to prevent ISIS LSP flooding.

Post-Check Verification

```
Command: show ip bgp, Expected Output: BGP table version is 12, main routing table version is 12, Description: This command verifies that the BGP table version and main routing table version are correct after clearing BGP route churn.

Command: show ip ospf interface, Expected Output: OSPF interface is in state UP, Description: This command verifies that the OSPF interface state is correct after re-establishing OSPF adjacency.

Command: show isis database, Expected Output: ISIS LSP flooding not detected, Description: This command verifies that ISIS LSP flooding is not detected after configuring ISIS parameters.
```

Roll-back Procedures

- Step: Restore previous BGP configuration, Command: copy running-config startup-config, Description: This command restores the previous BGP configuration from the startup configuration file, which can help resolve issues caused by the previous fix.

- Step: Re-establish OSPF adjacency, Command: clear ip ospf process, Description: This command re-establishes OSPF adjacency, which can help resolve issues caused by the previous fix.

- Step: Re-configure ISIS parameters, Command: router isis, Description: This command re-configures the ISIS protocol and sets the necessary parameters to prevent ISIS LSP flooding, which can help resolve issues caused by the previous fix.